The Effect of Expected Income on Individual Migration Decisions

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Roadmap

Roadmap

Models of Migration

- Migration models are used to estimate how and why agents move within an economy
- Typically used to look at isolated migration events, which may be too shallow of a viewpoint

Contribution (Short Discussion)

- **What's new**: Develops a tractable econometric model focusing on expected income's influence on migration which allows for dynamic decision making
- **Why it matters**: Adds possibility of both repeat and return migration, creating a context for this model to solve a job-search problem
- Using data from NLSY, gain an understanding of the motives behind and patterns exhibited by movers in the framework of this model

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Paper's Story / Key Findings

- The paper aims to look at white men with high school education, utilizing the NLSY to track agents through time
- Expected income prospects are a major contributor to interstate migration decisions
- Further, there is evidence of geographic factors taking place once income realization is met, and the current location is found to be suboptimal
- **Key quantitative takeaways**
  - Value of moving from a bad location to a good one equates to a $17,000 per year
  - Moving is mostly decided by fit of draw: if good, they stay; if bad: they try again, incurring a cost

Moving Statistics

TABLE I
INTERSTATE MIGRATION, NLSY 1979–1994[a]

	Less Than High School	High School	Some College	College	Total
Number of people	322	919	758	685	2,684
Movers (age 18–27)	80	223	224	341	868
Movers (%)	24.8%	24.3%	29.6%	49.8%	32.3%
Moves per mover	2.10	1.95	1.90	2.02	1.98
Repeat moves (% of all moves)	52.4%	48.7%	47.4%	50.5%	49.5%
Return migration (% of all moves)					
Home	32.7%	33.1%	29.1%	23.2%	28.1%
Not home	15.5%	7.1%	6.8%	8.6%	8.4%
Movers who return home	61.3%	56.5%	51.3%	42.8%	50.2%

[a]The sample includes respondents from the cross-section sample of the NLSY79 who were continuously interviewed from ages 18 to 28 and who never served in the military. The home location is the state of residence at age 14.

- 1.95 average lifetime moves
- 48.7% of movers will move again
- 56.5% of movers will return home

Roadmap

Intuition

Assume wages are a local product of skill bundles

Environment

- J locations
- y_{ij} is individual i's income in location j with a known distribution
- Migration occurs to maximize expected discounted value of lifetime utility
- Ignore assets for this case, only seeking to maximize expected lifetime income, net of moving costs

Value Function

$$V(x,\zeta) = \max_j (v(x,j) + \zeta_j) \tag{1}$$

where,

$$v(x,j) = u(x,j) + \beta \sum_{x'} p(x'|x,j)\bar{v}(x')$$

and

$$\bar{v}(x) = \mathbb{E}_\zeta V(x,\zeta)$$

Value Function, cont.

Further,

$$\exp(\bar{v}(x)) = \exp(\bar{\gamma}) \sum_{j=1}^{J} \exp(v(x,k)),$$

and

$$\rho(x,j) = \exp(\bar{\gamma} + v(x,j) - \bar{v}(x))$$

where $\bar{\gamma}$ is the Euler constant, $\rho(x,j)$ is the probability of choosing j when in state x

Wages

Wage of agent i in location j at age a during time t is as follows,

$$w_{ij}(a) = \mu_j + v_{ij} + G(X_i, a, t) + \eta_i + \epsilon_{ij}(a) \tag{2}$$

- Quality of match governed by the difference between $\mu_j + v_{ij}$ and $\mu_k + v_{ik}$, the current quality versus the prospective quality of another location
- Match quality is known for current and, where possible, most recent location
- Prospects of all other locations are unknown

State Variables & Flow Payoffs

- $l = (l^0, l^1, ..., l^{M-1})$ be an M vector containing the sequence of previous locations, beginning with the current location
- ω is an M vector containing wage and utility information at these locations
- $l, \omega, a \in x$, the state vector

$$\tilde{u}_h(x, j) = u_h(x, j) + \zeta_j, \tag{3}$$

where,

$$u_h(x, j) = \alpha_0 w(l^0, \omega) + \sum_{k=1}^{K} \alpha_k Y_k(l^0) + \alpha^H(l^0 = h) + \xi(l^0, \omega) - \Delta_\tau(x, j)$$

Moving Costs

- $D(l^0, j)$ represents distance from current location to some other location j
- $\mathbb{A}(l^0)$ set of locations adjacent to l^0

Thus, define moving cost as

$$\begin{aligned}\Delta_\tau(x, j) = (&\gamma_{0,\tau} + \gamma_1 D(l^0, j) - \gamma_2(j \in \mathbb{A}(l^0)) \\ &- \gamma_3(j = l^1) + \gamma_4 a - \gamma_5 n_j)(j \neq l^0)\end{aligned} \tag{4}$$

Transition Probabilities

$$p(x'|x,j) = \begin{cases} 1, & \text{if } j = l^0, \tilde{x}' = \tilde{x}, a' = a+1, \\ 1, & \text{if } j = l^1, \tilde{x}' = (l^1, l^0, x_v^1, x_v^0, x_\xi^1, x_\xi^0), a' = a+1, \\ \frac{1}{n^2}, & \text{if } j \notin \{l^0, l^1\}, \tilde{x}' = (j, l^0, s_v, x_v^0, s_\xi, x_\xi^0), \\ 0, & \text{otherwise} \end{cases}$$

such that,

- $x = (\tilde{x}, a)$
- $\tilde{x} = (l^0, l^1, x_v^0, x_v^1, x_\xi^0, x_\xi^1)$

Roadmap

Data: Sources & Coverage

Data Sources

- NLSY79: annual interviews from 1979-1994
- Supplemented with PUMS data - 1990 Census

Subset

- White (non-hispanic)
- High school graduates with no post-secondary education
- Years after schooling completed
- Age 20 until 1994, or the first year in which there are relevant missing data points

Data Use (for Each Data Type)

PUMS data

- Used for state mean effects
- White high school men aged 19-20
- Estimate means with median regression employing age and state dummies

NLSY

- Gain wage information
- Gain employment information
- 432 units with 4,274 person years

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Likelihoood Function

$$\Lambda(\theta) = \sum_{n=1}^{N} \log(\sum_{\tau=1}^{K} \pi_\tau L_i(\theta_\tau)) \tag{5}$$

- Observed income
- Location choice

Distributions and Definitions

In each location there is a draw from the distribution of location match wage components

$$\Upsilon = \{\upsilon(1), \upsilon(2), ..., \upsilon(n_\upsilon)\} \text{ indexed by } \omega_\upsilon \tag{6}$$

$\omega_\upsilon(j)$ is the matching component in location j s.t. $1 \leq \omega_\upsilon(j) \leq n_\upsilon$
There is also a location match preference distribution which is modeled as uniform over the finite set

$$\Xi = \{\xi(1), \xi(2), ..., \xi(n_\xi\}) \text{ indexed by } \omega_\xi \tag{7}$$

Distributions and Definitions, cont.

There is a uniform distributino of fixed effects they draw from

$$H = \{\eta(1), \eta(2), ..., \eta(n_\eta\}) \text{ indexed by } \omega_\eta \tag{8}$$

Finally, there is a draw from a uniform transient variance distribution given as

$$\zeta = \{\sigma_\epsilon(1), \sigma_\epsilon(2), ..., \sigma_\epsilon(n_\epsilon) \text{ indexed by } \omega_\epsilon \tag{9}$$

Distributions and Definitions, cont.

Wages and Preferences:

- Unobserved components of wages and preferences for each agent are given by vector $\omega^i = \{\omega^i_\xi, \omega^i_\eta, \omega^i_\epsilon, \omega^i_\upsilon(1), \omega^i_\upsilon(2), ..., \omega^i_\upsilon(n_\eta)$ which has $N_i + 3$ elements
- N_i is number of locations visited by the individual i
- Set of possible realizations of ω^i is $\Omega(N_i)$, which has $n_\xi n_\eta n_\epsilon (n_\upsilon)^{N_i}$ points

Location Indexing:

- κ^0_{it}, κ^1_{it} represent position of current and previous locations
- $\kappa_{it} = (\kappa^0_{it}, \kappa^1_{it})$ is a pair of integers betweeen $1, N_i$
- i.e. someone who never moves: $\kappa_{it} = (1, 0)$

Likelihoods

Let $\psi_{it}(\omega_i, \theta)$ correspond to likelihood of observed income of agent i in period t

$$\begin{aligned}\psi_{it}(\omega_i, \theta_\tau) \\ &= \frac{1}{\sigma_\epsilon(\omega^i_\epsilon)}\phi\Big(\frac{w_{it} - \mu_{l^0(i,t)}0G(X_i, a_{it}, \theta_\tau) - v(\omega^i_v(\kappa^0_{it})) - \eta(\omega^i_\eta)}{\sigma_\epsilon(\omega^i_\epsilon)}\Big)\end{aligned} \tag{10}$$

Let $\lambda_{it}(\omega^i, \theta_\tau)$ be the likelihood of the destination chosen by agent i in period t

$$\begin{aligned}\lambda_{it}(\omega^i, \theta_t) =& \rho_{h(i)}(l(i,t), \omega^i_v(\kappa^0_{it}), \omega^i_v(\kappa^1_{it}), \omega^i_\xi(\kappa^0_{it}), \\ & \omega^i_\xi(\kappa^1_{it}), a_{it}, l^0(i, t+1), \theta_\tau)\end{aligned} \tag{11}$$

Likelihoods, cont.

Likelihood of individual history for a person of type τ is

$$L_i(\theta_\tau) = \frac{1}{n_\eta n_\epsilon n_\xi (n_\upsilon)^{N_i}} \times \sum_{\omega^i \in \Omega(N_i)} \left(\Pi_{t=1}^{T_i} \psi_{it}(\omega^i, \theta_\tau) \lambda_{it}(\omega^i, \theta_\tau)\right) \tag{12}$$

Summary Statistics & Balance

Table: Summary / Balance (Illustrative)

Variable	Control	Treat	Diff
Outcome (baseline)			
Covariate A			
Covariate B			

Treatment vs. Control (Definition)

- Operational definition of **treatment** and **control**.
- Timing / exposure, staggered adoption if any.
- Sample restrictions and measurement windows.

Alternative Specifications — Preview

- Heterogeneity (hospital size, region, worker type).
- Functional form variants / additional controls.
- Event study / placebo / robustness roadmap.

Roadmap

Main Results: Table

Table: Main Specification (Illustrative)

	(1)	**(2)**	**(3)**	**(4)**
Treat				
Controls				
FE				
N				
R^2				

:

Side-by-Side Comparisons

Roadmap

Robustness / Alternative Specs

- Sensitivity to windows / bandwidths.
- Alternative outcomes / subsamples.
- Placebo and permutation checks.

Roadmap

Conclusion

- **What we learned** — 2–3 bullets.
- **Policy/economic implications**.
- **Limitations** and next steps.

Thank You

Questions?

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Data Details

- Construction and variable definitions.
- Additional balance / coverage checks.